

Response to Reviewers

Manuscript: XHBot: eXplainable Heterophily-aware Graph Neural Networks for Social Bot Detection

Journal: EAI Endorsed Transactions on AI and Robotics

Decision: Revisions Required

Dear Associate Editor and Reviewers,

We thank the Associate Editor and both reviewers for their careful and constructive evaluation of our manuscript. The comments have substantially improved the clarity, rigour, and presentation of the work. We have revised the manuscript thoroughly and respond to every point below. Throughout this letter we use the following convention: reviewer comments appear in *shaded italic boxes*, our responses follow in regular text, and the most relevant added or modified manuscript text is reproduced in **green-ruled boxes**. All changes compile cleanly; the revised LaTeX source and the compiled PDF accompany this letter.

Summary of the principal changes

1. Removed the duplicated Conclusion section and the inconsistent “+27.37%” figure; the manuscript now reports a single, consistent set of improvements (9.64% on TwiBot-20, 13.15% on TwiBot-22).
2. Rewrote the abstract as a single flowing paragraph without structured labels.
3. Added a spectral-theory justification for SGTR based on the graph Dirichlet energy, and an explicit comparison with CARE-GNN and RABot.
4. Rebuilt the ablation study to ablate each of the three THCA channels independently and to include *w/o CPD* and *w/o SGTR* variants; corrected the erroneous “Dual-Channel” terminology.
5. Added a baseline-fairness statement (identical splits, identical imbalance sampling across all models, re-tuned hyperparameters).
6. Added a qualitative forensic case study for the HFE module.
7. Replaced the cluttered architecture figure with a clean vector diagram that explicitly shows the three THCA channels; clarified the robustness figure and expanded its analysis.
8. Softened absolute/exaggerated language throughout; standardised the bibliography (DOIs, author lists) and in-text citation style; and specified the previously missing hyperparameters.

Response to Reviewer 1

Comment 1 — Novelty of SGTR relative to CARE-GNN and RABot.

One of the core modules, “Spectral-Guided Topology Refinement (SGTR),” aims to prune camouflage edges by calculating edge spectral energy. However, this concept is highly similar in both motivation and implementation to existing works, such as the label-aware similarity metric in CARE-GNN and the reinforcement learning-guided edge filtering in RABot. ... Is merely replacing the similarity metric with “spectral energy” sufficient to constitute a core innovation?

Response. We agree the original text did not adequately differentiate SGTR. We have (i) added a spectral-theory derivation grounding the “spectral” terminology (see Reviewer 2, Comment 3),

and (ii) added a dedicated paragraph, “*Relation to prior edge-refinement methods*,” in Section 3.2, and a contrasting sentence in the related-work RABot paragraph. We now position SGTR as a label-free, theoretically motivated, single-step differentiable alternative, and we measure its isolated contribution in the ablation (Table 2, *w/o SGTR*, -0.0494 F1).

Revised text: *SGTR is related to, but distinct from, two lines of prior work. CARE-GNN identifies informative neighbours using a label-aware similarity measure trained with node labels and a reinforcement-learning neighbour selector; RABot discards edges below a reliability threshold that is itself tuned online by a separate reinforcement-learning agent. In contrast, SGTR (i) is label-free, scoring edges purely from feature discrepancy and thus avoiding the propagation of label noise into the topology; (ii) is grounded in the graph Dirichlet-energy decomposition, giving the score a spectral interpretation rather than a heuristic similarity; and (iii) is a single differentiable pre-aggregation step trained end-to-end with the detector, rather than a separately optimised RL policy.*

Comment 2 — Baseline fairness and imbalance sampling.

...the paper mentions that XHBot uses “Imbalance Sampling” to handle class imbalance. The reviewer questions whether this strategy was applied to all baseline models as well. If the baseline models were not evaluated using the same data sampling strategy, the performance improvements might be partially attributed to data processing techniques rather than the inherent superiority of the model architecture.

Response. This is an important point. We added Section 4.1.4, “Baseline Implementation and Fair Comparison,” which states that imbalance-aware sampling was applied uniformly to all baselines that support it, that all models share identical splits/features/graph, and that baseline hyperparameters were re-tuned per dataset. The *w/o Imbalance Sampling* ablation row isolates this component for XHBot.

Revised text: *...the class-imbalance handling used by XHBot (imbalance-aware sampling) was applied uniformly to all baselines that support it, so no model receives an advantage from sampling alone... For each baseline we started from the authors’ reference implementation and hyperparameters where available, and additionally re-tuned the key hyperparameters ... on the validation set of each dataset. Consequently, the comparison reflects re-optimised baselines on the datasets used in this study rather than figures transcribed from their original papers.*

Comment 3 — Ablation study flaws (“Dual-Channel” vs. Tri-channel; missing CPD).

The ablation study (Table 2) has obvious flaws. For example, it removes “Dual-Channel Aggregation,” whereas the THCA module is actually “Tri-channel.” ... Furthermore, the paper lacks an ablation study for the “Contrastive Prototype Disentanglement (CPD)” module ... Table 2 does not include a “w/o CPD” variant.

Response. We rebuilt Table 2 entirely. It now ablates each THCA channel independently (*w/o Homophilic*, *w/o Heterophilic*, *w/o Self-Identity*), retains a homophilic-only extreme for reference, and adds *w/o SGTR* and *w/o CPD*, with a new Δ F1 column. The erroneous “Dual-Channel” label has been removed throughout. Removing the heterophilic channel produces the largest single-channel drop (-0.1244 F1); *w/o CPD* quantifies the disentanglement contribution (-0.0424 F1), showing CPD improves separability beyond THCA alone.

Comment 4 — HFE explainability is evaluated only quantitatively.

...metrics like Fidelity and Sparsity ...are far from sufficient for a human-oriented task like “forensic-level explanation.”

Response. We added Section 4.6, “Qualitative Forensic Case Study,” which walks a representative flagged account through the three HFE evidence levels (instance subgraph, community ring, diagnostic motif) with a human-readable interpretation, and we explicitly acknowledge that automated metrics are necessary but not sufficient.

Revised text: *...these three levels turn an opaque probability into a reviewable narrative (“this account, together with these 11 others, follows many unrelated humans in a coordinated star pattern”), which is the form of evidence a moderation team can act on and document. We emphasise that this case study is illustrative; a rigorous assessment of whether such explanations improve human moderator decisions would require a dedicated user study, which we identify as future work.*

Comment 5 — Inconsistent “+27.37%” claim in the conclusion.

In the conclusion (Section 7), the paper claims “securing a massive +27.37% F1 score improvement.” This figure is completely inconsistent with the “+9.64%” and “+13.15%” reported in the main experimental section (Section 4.2).

Response. The inconsistency arose from an accidentally duplicated Conclusion section. We deleted the duplicate section and the erroneous figure. The manuscript now reports only the values supported by Section 4.2 (9.64% on TwiBot-20, 13.15% on TwiBot-22), stated consistently in the abstract, introduction, results, and conclusion.

Comment 6 — Figure clarity (architecture and robustness).

As the model architecture diagram, Fig. 1 is highly informative but somewhat cluttered ... the internal details of certain modules (e.g., the three channels of THCA) are not clearly expressed. The title of Fig. 4 is “Model robustness ...” but the specific meanings of the curves ... are unclear in the legend, and the analysis of this figure in the main text is rather brief.

Response. Figure 1 was redrawn as a clean vector (TikZ) diagram with an explicit left-to-right data flow and a dedicated, labelled THCA block showing all three channels (homophilic / heterophilic / self-identity). Figure 4 (robustness) was regenerated with a titled legend, an explicit x-axis definition of “camouflage degree,” distinct markers, and end-point value annotations; its analysis was expanded from a single sentence to a full paragraph quantifying each model’s degradation.

Comment 7 — Exaggerated language.

The paper frequently uses absolute terms such as “fundamentally insufficient” and “impenetrable black boxes” ... it is recommended to use more objective and neutral language.

Response. We revised absolute phrasing throughout, including “impenetrable black boxes” → “provide limited rationale,” “fundamentally insufficient” → “often insufficient,” “provably insufficient” → “often insufficient,” and “comprehensively dominates / massive margin / weaponizes” → neutral, quantitative descriptions.

Comment 8 — Reference inconsistencies (DOIs, author lists).

In the reference list, some entries include DOI links while others do not. The formatting of the author lists for some conference papers is also inconsistent.

Response. We standardised the bibliography. The arXiv entries now use a uniform `journal = {arXiv preprint} + doi` format with redundant URL/identifier fields removed; a DOI was added to the GNNExplainer entry; and author lists use a uniform `Last, First` format. The two genuine no-DOI NeurIPS proceedings entries retain a canonical URL. We also fixed a BibTeX warning by removing a conflicting `number` field in the RGT entry.

Comment 9 — Optimise figures and language.

Response. Addressed jointly with Comments 6 and 7 above.

Response to Reviewer 2

Major comments

Major 1 — Abstract format.

The abstract uses explicit labels like “INTRODUCTION” and “METHODS.” This is non-standard for most journals. It should be rewritten as a single, flowing paragraph.

Response. The abstract has been rewritten as a single flowing paragraph; all structured labels have been removed.

Major 2 — Structure and redundancy (two Conclusions; verbose intro/related work).

...there are two separate “Conclusion” sections (Section 6 and Section 7) that repeat similar points. The authors must consolidate these ...the introduction and related work sections are overly verbose and could be significantly streamlined.

Response. The duplicated Conclusion was removed, leaving a single cohesive conclusion. We condensed the statistics-heavy opening of the introduction and substantially shortened the related-work section (notably the RABot and FEDRIO paragraphs) while preserving the technical contrasts.

Major 3 — Clarity of the SGTR module; “Spectral-Guided” is misleading.

...The text claims that cross-class connections manifest as “high-frequency noise,” yet Equation (1) defines a “spectral energy” ...using a learnable neural network on node features, not a spectral graph filter. The term “Spectral-Guided” is therefore misleading. ...How does learning an edge weight based on feature discrepancy relate to spectral graph theory?

Response. We added a “Spectral motivation” paragraph that makes the connection explicit. The graph Dirichlet energy (Laplacian quadratic form) decomposes over edges, so the squared feature discrepancy used in the score is exactly the per-edge contribution to the high-frequency (Dirichlet) energy; down-weighting high-energy edges therefore attenuates the high-frequency band that drives over-smoothing. The learnable projection is presented as a *calibrated, data-dependent estimate* of this quantity, which justifies the terminology.

Revised text: For a graph signal $\mathbf{f} \in \mathbb{R}^N$ and the combinatorial Laplacian $L = D - A$, the Dirichlet energy admits the edge decomposition $\mathbf{f}^\top L \mathbf{f} = \sum_{(u,v) \in \mathcal{E}} (f_u - f_v)^2$, which measures the high-frequency content of $\mathbf{f}$ the squared feature discrepancy $(\mathbf{x}_u - \mathbf{x}_v)^2$ is the per-edge contribution to the total feature Dirichlet energy. ... SGTR scores edges by this quantity and suppresses the highest-energy ones, which is equivalent to attenuating the high-frequency band that drives over-smoothing in low-pass GCN aggregation.

Major 4 — Justification for the CPD module (add “w/o CPD”).

... The ablation study in Table 2 ... does not include a variant “w/o CPD.” The authors must add this ablation ... Without it, it is unclear if the performance gains are driven by the THCA module alone.

Response. Added (see Reviewer 1, Comment 3). Table 2 now includes *w/o CPD* (−0.0424 F1), demonstrating that CPD contributes beyond THCA alone, primarily by sharpening class separability.

Major 5 — Explainability evaluation: add qualitative case studies.

... The authors should include qualitative case studies. For example, show a specific bot account flagged by the model, present the subgraph identified by HFE as the “cause,” and provide a human-interpretable explanation of why that subgraph is suspicious ...

Response. Added (see Reviewer 1, Comment 4). Section 4.6 presents a concrete flagged account, the HFE subgraph as the cause, and a human-interpretable explanation of the “star-formation” botnet pattern, exactly as suggested.

Major 6 — Baseline selection and fairness.

... It would strengthen the paper to discuss whether the hyperparameters for these baseline models were re-optimized on the specific datasets used in this study, or if the results were taken from their original papers. Ensuring a fair comparison is critical.

Response. Addressed in Section 4.1.4 (see Reviewer 1, Comment 2): baseline hyperparameters were re-tuned on this study’s datasets rather than transcribed from the original papers, and the full comparison protocol is now stated explicitly.

Minor comments

Minor 1 — Figure 1 density.

Response. The architecture diagram was redrawn as a clean vector figure with a clearer data flow and an explicit THCA grouping (see Reviewer 1, Comment 6).

Minor 2 — Missing hyperparameters (hidden dimension, number of layers).

... does not specify the hidden dimension size or the number of layers for the GNN backbone. These details are necessary for reproducibility.

Response. Section 4.1.3 now specifies the number of THCA layers ($L = 2$), hidden dimension ($d_h = 64$), dropout (0.5), CPD sub-space dimension ($d' = 32$), InfoNCE temperature ($\tau_{nce} = 0.1$), loss weights ($\lambda_{MI} = 0.1$, $\lambda_{NCE} = 0.5$), and the SGTR threshold ($\tau = 0.5$).

Minor 3 — Confusing notation in Equation (2).

Typo in Equation (2): the notation is slightly confusing. Please clarify.

Response. The soft-pruning rule was rewritten in explicit piecewise (cases) form, making the thresholding behaviour unambiguous: low-energy edges keep their original weight, while high-energy edges are attenuated by a factor $(1 - E)$.

Revised text: $\tilde{A}_{u,v}^{(r)} = A_{u,v}^{(r)} (1 - E^{(r)}(u, v))$ if $E^{(r)}(u, v) > \tau$, and $\tilde{A}_{u,v}^{(r)} = A_{u,v}^{(r)}$ otherwise.

Minor 4 — Citation formatting.

There are some inconsistencies in the citation style (e.g., “[17] proposes MGDIL” vs “The work of [17] proposes...”). Please standardize the citation format throughout.

Response. We standardised in-text citations to the “Author et al. [n]” form when a work is the subject of a sentence (e.g., “Qiao et al. [17] propose MGDIL,” “Fu et al. [8],” “He et al. [9],” “Dang and Nguyen [15]”), removing the inconsistent bare-number and “The work of . . .” constructions.

We believe these revisions comprehensively address the reviewers’ concerns and meaningfully strengthen the manuscript. We thank the Associate Editor and both reviewers again for their time and valuable feedback.

Sincerely,
The Authors